Free-Energy Inference for Verifier-Guided Structured Generation

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Planning draft

Abstract

This paper is the conditional upside paper for the Free Energy project. It should be advanced only if the proposed novel EBM produces a concrete advantage over autoregressive and conventional EBM baselines on verifier-guided structured generation.

1 Problem

Autoregressive models turn global structured generation into local next-token prediction. Energy-based models score whole objects but inherit hard normalization and sampling. The goal of this paper is to propose and test a model that keeps the global compatibility benefits of energies while making training and inference practical enough for verifier-guided tasks.

2 Model

The model should be stated as an energy or free-energy objective. At minimum, specify:

$$E_\theta(x, y) \tag{1}$$

for context x and candidate structured object y, plus the training surrogate that avoids or approximates

$$Z_\theta(x) = \sum_y \exp(-E_\theta(x, y)). \tag{2}$$

3 Training objective

The paper should state exactly which cost is being avoided or relocated:

- local softmax normalization;
- global partition-function estimation;
- verifier-guided contrastive or margin loss;
- free-energy/variational surrogate;
- test-time energy descent or search.

4 Inference

Inference should be described as a budgeted optimization procedure rather than a vague global search. Report the number of energy evaluations, verifier calls, gradient steps, samples, and retries.

5 Experiments

1. Toy controls where the mechanism is known to help.
2. VeriBench/Lean structured generation.
3. MNIST/order-effect smoke if it clarifies the mechanism.
4. Ablations against AR and conventional EBM baselines.

6 Go/no-go criteria

This paper should not be submitted unless at least one of the following is true:

- pass@k improves over both AR and normal EBM baselines at matched budget;
- cost per verified solution improves substantially;
- scaling with proof/program difficulty is better;
- a negative result reveals a publishable failure mode of existing EBM assumptions.